

Date: 7/1/2025

Concepts

Thermodynamic equilibrium

Walls

Internal energy (U)

Postulate - ①

Isolated system

Closed system

Intensive vs Extensive

Explanation

• state at which all measured coordinates are time independent

• place restriction on change of one or more of these components

• Intensive (depends on amount of matter)

• measured in relation to another

• not dependent on how

• Existence of equilibrium

• adiabatic

• $\times$ energy/matter can cross

• diathermal

• $\times$ matter can cross

• dependent/independent of size of system

Notes

Date: 9/10/25

Concepts

Conjugate pair

Work

Heat

Postulate (2)

• Postulate (3)

Zeroth law

Explanation

• Change in X leads to change in Y (P & V)

• $-\int P dv$ - for quasistatic process

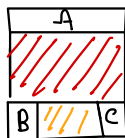
• $dQ = dU - dW$

• @ equilibrium entropy (S) is max $\therefore dS = 0$

• @ $T = 0$; $S = 0$

• $\frac{dU}{dS} = 0 \Rightarrow T = 0$

•



→ diathermal wall

→ adiabatic wall

if AB & AC are in equilibrium
 BC are in equilibrium

Notes

quasistatic $\rightarrow$ infinitely slow, always in equilibrium
work to expand to vacuum is 0

Date: 10/01/2025

Concepts

Temperature

1st law of Thermo.

2nd law

3rd law

Conditions for equilibrium

State function

Path variables

Explanation

- defines an equilibrium
- every system in equilibrium is in same temperature
- conservation of energy
- $dQ = dU - dW$
- Σ entropy never decreases
- $T \rightarrow 0; S \rightarrow 0$
- Thermal equilibrium (T)
- Mechanical equilibrium (P)
- Chemical equilibrium (μ)
- relates 2+ state variables
- depend on path taken to achieve the state unlike state variables

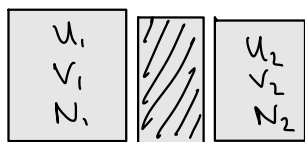
Notes

Date:

Derivation of equilibrium condition

$$S = S(U, V, N)$$

For a composite system $U, V, N \rightarrow$ conserved



$$\Rightarrow U_1 + U_2 = C \Rightarrow dU_1 + dU_2 = 0$$

$$\Rightarrow V_1 + V_2 = C \Rightarrow dV_1 + dV_2 = 0$$

$$\Rightarrow N_1 + N_2 = C \Rightarrow dN_1 + dN_2 = 0$$

$\hookrightarrow$ wall permeable to molecules, diathermal, & movable

$$S = S_1 + S_2$$

@ equilibrium

$$dS = dS_1 + dS_2 = 0$$

$$\Rightarrow \frac{\partial S_1}{\partial U_1} dU_1 + \frac{\partial S_1}{\partial V_1} dV_1 + \frac{\partial S_1}{\partial N_1} dN_1 + \frac{\partial S_2}{\partial U_2} dU_2 + \frac{\partial S_2}{\partial V_2} dV_2 + \frac{\partial S_2}{\partial N_2} dN_2 = 0$$

$$\Rightarrow \left(\frac{\partial S_1}{\partial U_1} - \frac{\partial S_2}{\partial U_2} \right) dU_1 + \left(\frac{\partial S_1}{\partial V_1} - \frac{\partial S_2}{\partial V_2} \right) dV_2 + \left(\frac{\partial S_1}{\partial N_1} - \frac{\partial S_2}{\partial N_2} \right) dN_1 = 0$$

$$\Rightarrow \frac{\partial S_1}{\partial U_1} = \frac{\partial S_2}{\partial U_2} \Rightarrow \boxed{T_1 = T_2} \quad \left[\frac{\partial S}{\partial U} = \frac{1}{T} \right]$$

$$\Rightarrow \frac{\partial S_1}{\partial V_1} = \frac{\partial S_2}{\partial V_2} \Rightarrow \frac{\partial S_1}{\partial U_1} \frac{\partial U_1}{\partial V_1} = \frac{\partial S_2}{\partial U_2} \frac{\partial U_2}{\partial V_2} = \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \boxed{P_1 = P_2}$$

$$\Rightarrow \frac{\partial S_1}{\partial N_1} = \frac{\partial S_2}{\partial N_2} \Rightarrow \frac{\mu_1}{T_1} = \frac{\mu_2}{T_2} \Rightarrow \boxed{\mu_1 = \mu_2}$$

$\mu \rightarrow$ chemical potential

Date:

Concepts

Equation of state

Entropy fundamental

Energy fundamental

Fundamental relation

Energetic intensive quantities

Explanation

- Gives full info about system
- Expresses intensive variable in terms of measurable extensive variables
- Zeroth order ($\lambda E(x, y, z) \neq E(\lambda x, \lambda y, \lambda z)$)
- derivation leads to expression of states that can't be measured
- $S = (U, V, N, \dots)$
- $U = (S, V, N, \dots)$
- homogeneous first order equation of extensive variables
- $\Rightarrow \lambda F(a, b, c) = F(\lambda a, \lambda b, \lambda c)$
- derived from energy fundamental relations
- $T, P, \mu \rightarrow$ conditions for equilibrium

Notes

Date:

EOS - ①

$$V = V(T, P)$$

$$dV = \left(\frac{\partial V}{\partial T} \right) dT + \left(\frac{\partial V}{\partial P} \right) dP$$

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right) \quad \text{Thermal expansion coeff}$$

$$K_\beta = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad \text{Isothermal compressibility } (1/V_{\text{Bulk}})$$

$$dV = V\beta dT - VK_\beta dP$$

EOS of hydrostatic system

$$U = U(T, V)$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$dQ = dU - dW$$

$$= \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV + PdV$$

$$\left(\frac{dQ}{dT} \right)_V = C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$\left(\frac{dQ}{dT} \right)_P = C_P = C_V + \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right] \frac{dV}{dT}$$

$$\Rightarrow C_P - C_V = \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right] V\beta$$

$$\Rightarrow \left(\frac{\partial U}{\partial V} \right)_T = \frac{C_P - C_V}{V\beta} - P$$

Date:

Derivation of Energetic intensive quantities EOS

$$\bullet dU = \left(\frac{\partial U}{\partial S}\right) dS + \left(\frac{\partial U}{\partial V}\right) dV + \left(\frac{\partial U}{\partial N_i}\right) N_i$$

$$= T dS + P dV + \sum \mu_i N_i$$

If N_i is constant

$$\bullet dU = T dS - P dV \text{ [First law]}$$

$$\Rightarrow T dS = dQ$$

$$\bullet T = \frac{\partial U}{\partial S}$$

$$\Rightarrow T = T(S, V, N_1, \dots)$$

Similarly

$$P = P(S, V, N_1, \dots)$$

$$\mu = \mu(S, V, N_1, \dots)$$

Date: 16/01/2025

Concepts

Theorem

Euler's Equation

Gibbs Duhem Relation

Explanation

If (x, y, z) are thermodynamical variables @ there exists a function $f(x, y, z) = 0$ for all x, y, z

① $(\partial x / \partial y)_z = 1 / (\partial y / \partial x)_z$ derived by differentiating $x = x(y, z)$ & $y = y(x, z)$

② $(\partial x / \partial y)_z (\partial y / \partial z)_x (\partial z / \partial x)_y = -1$

• Establishes relation btwn different extensive variables

• $TS - PV + \sum \mu_i N_i = U$

• derived from differentiating

$U(\lambda S, \lambda V, \dots) = \lambda U(S, V, \dots)$ wrt λ

• From derivation of Euler's Equation

• relationship between different intensive variables

• $SdT - VdP + \sum N_i d\mu_i = 0$

Notes

Date: 19/01

Derivation of fundamental equation for ideal gas!

$$\textcircled{1} pV = NRT \Rightarrow p/T = NR/v$$

$$\textcircled{2} u = cRT \cdot N \Rightarrow v/T = cNR/u$$

$$\textcircled{3} \mu/T = ? \quad u d(1/T) + v d(p/T) - N d(\mu/T) = 0$$

$$\Rightarrow d(\mu/T) = u c R d(1/u) + R d(v/v)$$

$$\Rightarrow \frac{\mu}{T} - \frac{\mu_0}{T_0} = u c R \ln\left(\frac{u}{u_0}\right) + v R \ln\left(\frac{v}{v_0}\right) \rightarrow a$$

$$S = \frac{1}{T} u + \frac{p}{T} v - \frac{\mu}{T} N$$

$$= cRT + NR - [a] \cdot N$$

$$= N \left[(c+1)R + \mu_0/T_0 \right] + NR \ln \left[\left(\frac{u}{u_0}\right)^c \left(\frac{v}{v_0}\right) \left(\frac{N_0}{N}\right)^{c+1} \right]$$

$$ds = \frac{du}{T} + \frac{p dv}{T}$$

$$ds = cR \frac{du}{u} + R \frac{dv}{v}$$

$$s - s_0 = cR \ln\left(\frac{u}{u_0}\right) + R \ln\left(\frac{v}{v_0}\right)$$

$$s = s_0 + cR \ln\left(\frac{u}{u_0}\right) + R \ln\left(\frac{v}{v_0}\right) \quad \times N$$

$$\Rightarrow \boxed{s_0 = (c+1)R + \mu_0/T_0}$$

Date: 21/01

Concepts

Thermodynamic
Configuration

Reversible vs irreversible

Explanation

- quantities that span thermodynamic configuration space
- Hypothetical and high dimensional
- Surface of plot of eqn. $\rightarrow$ hypersurface
- conditions for viable $\uparrow$ all pts are in eqi.
 $*(\partial U / \partial S) > 0$ i.e. for all S , U is unique

- $A \rightarrow B$: entropy at B because maximum for any state btwn A & B
- for each process 1/ more constants are relaxed.

- $\Delta S > 0$: irreversible
- $\Delta S = 0$: reversible
- $\Delta S < 0 \rightarrow$ not viable

If quasi-static

$$dW = -PdV$$

$$dQ = TdS$$

Notes

Date:

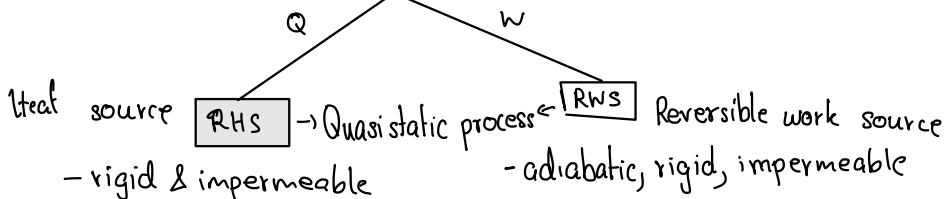
Maximum work theorem

→ For any reversible process $A \rightarrow B \rightarrow$ Subsystem

- work $\rightarrow \max$

- $Q \rightarrow \min$

$$\Delta U = U_B - U_A$$



→ From conservation of energy

$$-\Delta U = Q_{RHS} + W_{RWS}$$

$$-dU = dQ_{RHS} + dW_{RWS} \quad - (1)$$

→ Total change in entropy

$$0 = dS_{tot} = dS + \frac{dQ_{RHS}}{T_{RHS}}$$

$$\Rightarrow T_{RHS} \cdot dS + dQ_{RHS} = 0$$

→ Subbing (1)

$$T_{RHS} \cdot dS + (-dU - dW_{RHS}) = 0$$

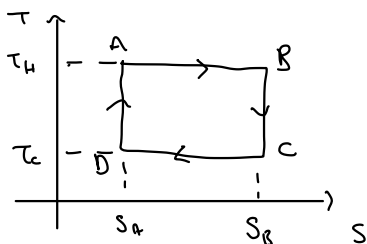
$$\Rightarrow dW_{RHS} = T_{RHS} \frac{dQ}{T} - dQ - dW$$

$$\Rightarrow dW_{RHS} = (-dQ) \left[1 - \frac{T_{RHS}}{T} \right] + (-dW)$$

$$\bullet \text{ Efficiency: } \left[1 - \frac{T_{RHS}}{T} \right]$$

Date:

Carnot cycle



$$A \rightarrow B: \text{Heat} = T \Delta S$$

$$= T_H (S_B - S_A) = T_H \Delta S$$

$$C \rightarrow D: \quad " \quad = T_C (S_A - S_B) = -T_C \Delta S$$

$$\text{net work} = T_H - T_C \Delta S$$

$$\eta = \frac{\text{output}}{\text{input}} = 1 - \frac{T_C}{T_H}$$

Efficiency of engine

$$dU + dQ_{RHS} + dW_{RHS} = 0$$

$$dU_h + dQ_c + dW_c = 0$$

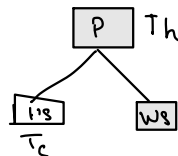
$$\Rightarrow dQ_h + dW_h + dW_c + dQ_c = 0$$

$$S_{\text{tot}} = dS_h + \frac{dQ_c}{T_c} = 0$$

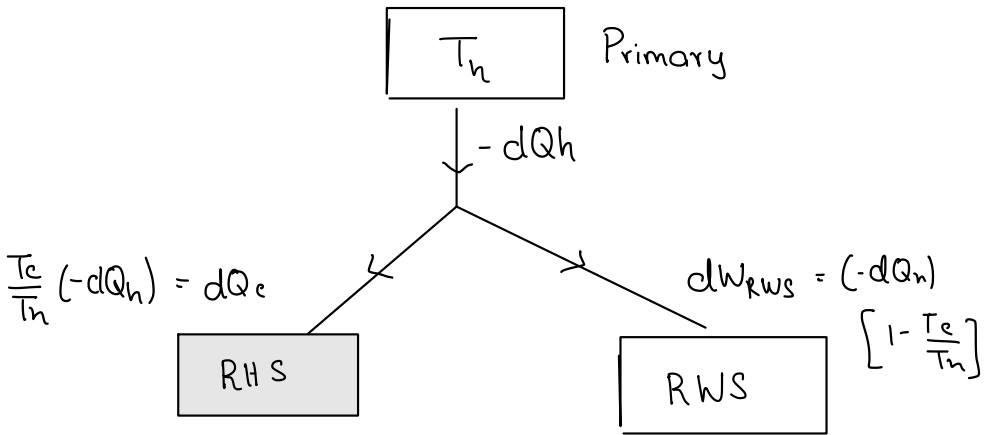
$$\Rightarrow T_c dS_h + dQ_c = 0$$

$$\text{Efficiency} = \frac{dW_{RHS}}{dQ_h} = \frac{dQ_h - dQ_c}{dQ_h}$$

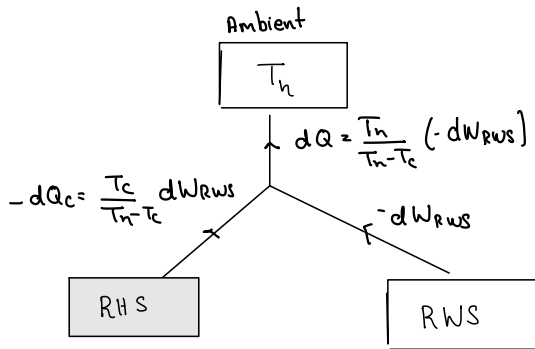
$$= \left[1 - \frac{dQ_c}{dQ_h} \right] = \left[1 - \frac{T_c}{T_H} \right]$$



Date: _____
Systems



Engine



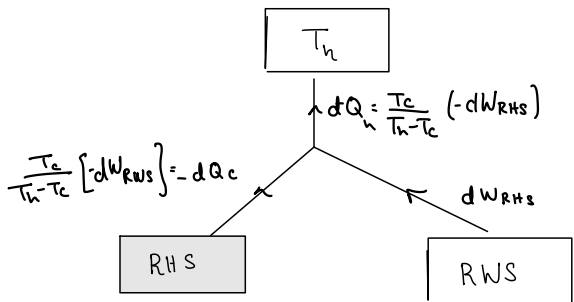
← Refrigerator

$$\epsilon = \frac{-dQ_c}{-dW_{RWS}}$$

$$= \frac{T_c}{T_h - T_c}$$

Heater

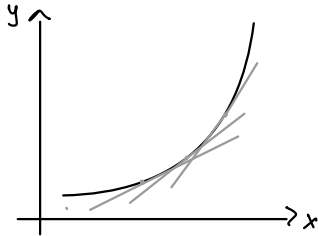
$$\eta = \frac{T_h}{T_h - T_c}$$



————— QUIZ-1 —————

Date: 28/01/2025

Legendre Transform



Let slope of tangent at
any point = p
& intercept = ψ

$$p = \frac{\partial y}{\partial x} \Rightarrow dy = p dx$$

$$= \frac{y - \psi}{x - 0}$$

$$\Rightarrow y = p x + \psi$$

$$\Rightarrow \psi = y - p x$$

$$(x, y) \rightarrow (p, \psi)$$

Both are fundamental
relations

$$\Rightarrow d\psi = \cancel{dy} - x dp - \cancel{p dx}$$

$$= -x dp$$

$$\Rightarrow \frac{d\psi}{dp} = -x$$

Original	Transform
$y = y(x)$	$\psi = \psi(p)$
$p = \partial y / \partial x$	$-x = d\psi / dp$
$y = \psi + p x$	$\psi = y - p x$

For $y = y(x_1, \dots, x_n)$

OC	L.T
$y = \psi + \sum p_k x_k$	$\psi = y - \sum x_k p_k$
$p_k = \frac{\partial y}{\partial x_k}$	$d\psi = -\sum x_k dp_k$

Date:

For partial Transform

$$y = y(x_1, \dots, x_n, \dots, x_t)$$

$\Rightarrow$ Partial derivative till x_n

$$y = y(p_1, p_2, \dots, p_n, x_{n+1}, \dots, x_t)$$

$$p_k = \left(\frac{\partial y}{\partial x_k} \right)$$

$Y[p_1, p_2, \dots, p_n]$: A fn of $p_1, \dots, p_n$ & $x_{n+1}, \dots, x_t$

$$\Rightarrow -x_k = \frac{\partial Y[p_1, \dots, p_n]}{\partial p_k} \quad k \leq n$$

$$\Rightarrow p_k = \frac{\partial Y(\quad)}{\partial x_k} \quad k > n$$

$$dy(\quad) = - \sum_1^n x_k dp_k + \sum_{n+1}^t p_k dx_k$$

Date: 30/01/2025

Thermodynamic Potential

→ Partial transfer of fundamental relation

$$U = U(S, V, N, \dots)$$

$$X_0 = S$$

$$P_0 = \partial U / \partial S = T$$

$$T dF = -S dT - P dV + \sum \mu_i dN_i$$

$$U[T] = U - TS = F \rightarrow \text{Helmholtz free energy}$$

Reservoir of T contact with complex system

$$X_0 = V$$

$$P_0 = -P$$

Min F $\rightarrow$ equi

$$U(P) = U - PV = H \rightarrow \text{Enthalpy} \quad dH = T dS + V dP + \sum \mu_i dN_i$$

$$\text{Both } U = U + PV - TS = G \quad \text{Gibbs}$$

$$dG = V dP - S dT + \sum \mu_i dN_i$$

$S = S(\) \rightarrow$ Transform: Massieu function

$$\bullet S[Y_T] = S - \frac{1}{T} U = F/T$$

$$\bullet S[P/T] = S + \frac{P}{T} V = H/T$$

$$\bullet S[P/T, Y_T] = S - \frac{P}{T} V - \frac{1}{T} U = G/T$$

Date:

Maxwells equation

$$\frac{\partial u}{\partial v \partial s} = \frac{\partial^2 u}{\partial s \partial v}$$

$$\Rightarrow \left(\frac{\partial T}{\partial v} \right)_{s,n} = \left(\frac{\partial p}{\partial s} \right)_{v,n}$$

$$\left(\frac{\partial T}{\partial n} \right)_{s,v} = \left(\frac{\partial \mu}{\partial s} \right)_{v,n}$$

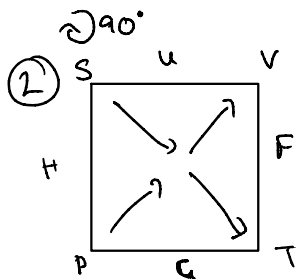
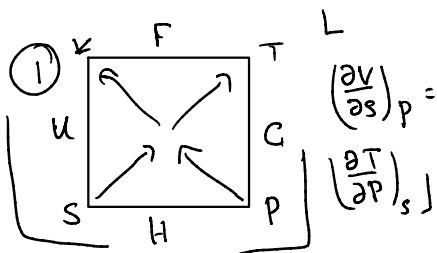
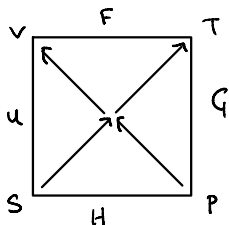
$$-\frac{\partial p}{\partial n} = \frac{\partial \mu}{\partial v}$$

Can be done for any state function

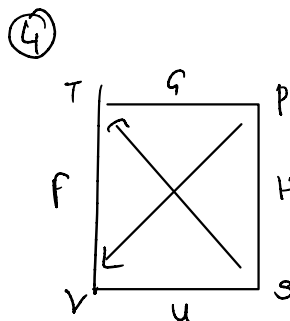
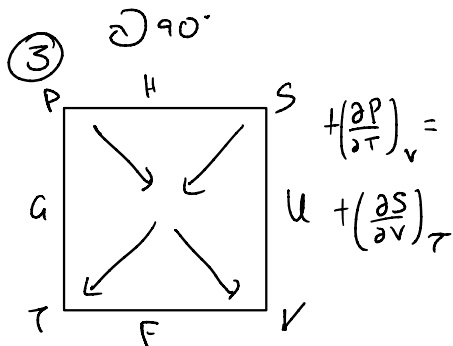
Verified facts and Through Understanding

Give Solutions to Hard Problems

Arrow away +ve towards -ve



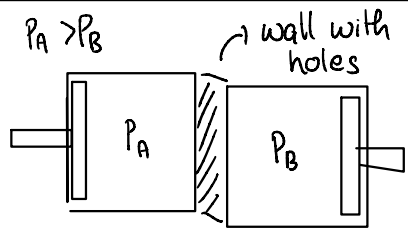
$$\left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p$$



$$-\left(\frac{\partial T}{\partial v} \right)_s = \left(\frac{\partial p}{\partial s} \right)_v$$

Date: 7/02/2025

JOULE THOMSON PROCESS



• closed system

$$H(T, P, N) = \text{const} \quad (\text{isoenthalpic})$$

$$\Rightarrow \partial T = \left(\frac{\partial T}{\partial P} \right) dP \quad \boxed{dN = 0}$$

$$= - \frac{(\partial H / \partial P)_T}{(\partial H / \partial T)_P} dP$$

$$* dH = T dS + V dP$$

$$\Rightarrow \partial T = - \frac{\left(\frac{\partial S}{\partial P} \cdot T + V \right)}{T \left(\frac{\partial S}{\partial T} \right)} dP$$

$$= - \frac{T \left(\frac{\partial V}{\partial T} \right)_P + V}{\left(\frac{\partial Q}{\partial T} \right)_P} dP$$

$$dT = \frac{V (\alpha T - 1)}{C_P} dP$$

For any $\psi(x, y, z) = 0$
 $@ d\psi = 0$

$$\left(\frac{\partial \psi}{\partial x} \right) dx + \left(\frac{\partial \psi}{\partial y} \right) dy + \left(\frac{\partial \psi}{\partial z} \right) dz = 0$$

$$dz = 0$$

$$\Rightarrow \left(\frac{\partial \psi}{\partial x} \right) dx + \left(\frac{\partial \psi}{\partial y} \right) dy = 0$$

$$\Rightarrow \frac{dx}{dy} = - \frac{\partial \psi / \partial y}{\partial \psi / \partial x}$$

For any gas generally $\alpha T > 1$

T' @ which $\alpha T < 1 \rightarrow$ inversion temp

$$T > T' \rightarrow dP < 0 \rightarrow \text{cools}$$

$$T < T' \rightarrow dP < 0 \rightarrow \text{warms up}$$

$$\mu_{JT} \text{ (JT's coeff)} = \left(\frac{\partial T}{\partial P} \right)_H$$

$$T' (O_2, N_2 \dots) = 623 \text{ K}$$

$$T' (H_2, He \dots) = 40 \text{ K}$$

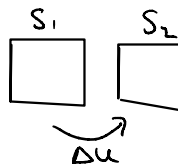
Date:

Conditions of stability

$$dS=0 [S \text{ max}], du=0 [u \text{ min}]$$

$$\text{OG/ } S_{1/2} = S(u, v, N); S_{\text{tot}} = 2S(u, v, N) - S$$

$$\text{Fin/ } S_{\text{tot}} = \overbrace{S(u+\Delta u, v, N)}^{S'_2} + \overbrace{S(u-\Delta u, v, N)}^{S'_1} - S' - \text{not on line}$$



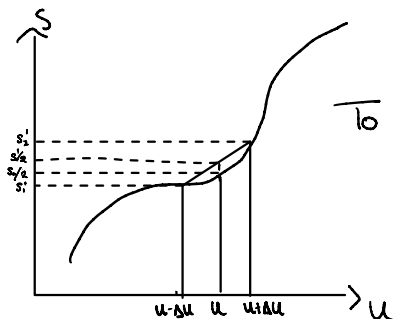
unstable equilibrium

$$S'_{1/2} > S/2$$

$$\text{To be stable } S'_1 + S'_2 \leq 2S$$

$$\text{when } \Delta u \rightarrow 0$$

$$\frac{\partial^2 S}{\partial u^2} \leq 0 \rightarrow \text{concave}$$



$$\text{If done for } \Delta V \Rightarrow \frac{\partial^2 S}{\partial v^2} \leq 0$$

Doing similar conditions we can prove that $\frac{\partial^2 u}{\partial s^2} \geq 0$ & $\frac{\partial^2 u}{\partial v^2} \geq 0$

ENTROPY (S)	ENERGY (U)
• Max @ Equilibrium • Convex function • $\frac{\partial^2 S}{\partial u^2} \leq 0$ • $\frac{\partial^2 S}{\partial v^2} \leq 0$	• Min @ equilibrium • Concave function • $\frac{\partial^2 u}{\partial s^2} \geq 0$ • $\frac{\partial^2 u}{\partial v^2} \geq 0$

Date:

Stability of Thermodynamic Potentials

From Legendre transform

$$p = dy/dx \quad \& \quad x = -dy[p]/dp$$

$$\partial x / \partial p = \partial^2 y[p] / \partial p^2 = -1 / \partial p / \partial x$$

$$\Rightarrow \partial^2 y[p] / \partial p^2 = -1 / \partial^2 y / \partial x^2$$

Taking $F(T, V, N)$

$$\partial^2 F / \partial T^2 = -1 / \partial^2 u / \partial s^2 \quad - \quad \partial^2 F / \partial V^2 \geq 0$$

$$\Rightarrow \partial^2 F / \partial T^2 \leq 0$$

Similarly

$$\frac{\partial^2 H}{\partial S^2} \geq 0 \quad ; \quad \frac{\partial^2 H}{\partial P^2} \leq 0$$

Extensive ≥ 0

Intensive ≤ 0

$$\left(\frac{\partial^2 G}{\partial T^2} \right) \leq 0 \quad \left(\frac{\partial^2 G}{\partial P^2} \right) \leq 0$$

Date:

Important consequences of condition of convexity!

$$\frac{\partial}{\partial u} \left(\frac{\partial S}{\partial u} \right) \leq 0$$

$$\Rightarrow \frac{\partial}{\partial u} \frac{1}{T} \leq 0$$

$$\Rightarrow -\frac{1}{T^2} \frac{\partial T}{\partial u} \leq 0$$

$$\Rightarrow -\frac{1}{T^2} \frac{1}{C_V} \leq 0$$

$$\Rightarrow T^2 C_V > 0$$

$$\Rightarrow C_V > 0 \quad [\text{for stable systems}]$$

$$\bullet C_P - C_V = \frac{T V \alpha^2}{\kappa_T}$$

$$\Rightarrow C_P - C_V > 0 \Rightarrow \boxed{C_P > C_V > 0}$$

$$\frac{\kappa_S}{\kappa_T} = C_V / C_P \Rightarrow \boxed{\kappa_T > \kappa_S \geq 0}$$

$$\bullet \partial^2 F / \partial V^2 \geq 0$$

$$\Rightarrow \frac{\partial}{\partial V} (-P) \geq 0$$

$$\Rightarrow + \frac{1}{V \kappa_T} \geq 0$$

$$\Rightarrow \kappa_T \geq 0$$

Date:

Gibbs Rule

For a system with r components and m phases in equilibrium

Total number of variables: $2 + (r-1)m$

Total number of equation: $(m-1)r$

$$f = V - E$$

$= r - m + 2$ - degrees of freedom

$$\geq 0$$

$\Rightarrow$ no of eqn $\geq$ no of variable for general solution

f : number of variables that can be arbitrarily assigned and still have an equilibrium exist!

Triple point (T/P) @ which all three states can coexist

for 1 component & 3 phases

$$f = 1 - 3 + 2 = 0$$

No variable can be assigned i.e. unique equilibrium

$$\left. \begin{aligned} S = S(T, V) &\Rightarrow Q = TdS = C_v dT + T(\partial p / \partial T)_V dV \\ S = S(T, P) &\Rightarrow TdS = C_p dT + T V \alpha dP \end{aligned} \right\} \text{2 eqns.}$$

Date:

First order phase transition

- $\text{solid} \rightleftharpoons \text{liquid} / \text{liquid} \rightleftharpoons \text{gas}$
- reversible
- @ transition boundary: all heat change goes to phase transition and not to temperature change
- Heat required for phase transition of one mole: latent heat

$$dh = Tds + vdp$$

$$\rightarrow dp = 0$$

$$L = Tds$$

- @ any moment x (mole fraction) substance is in final state

$$s = n_o(x) s^f + n_o(1-x) s^i$$

$$v = n_o x v^f + n_o(1-x) v^i$$

$$\begin{matrix} n_o \\ s^i \\ v^i \end{matrix}$$

$$\begin{matrix} n_o \\ s^f \\ v^f \end{matrix}$$

$$dg = s dT + v dp$$

$$s = \frac{\partial g}{\partial T}$$

$$v = \frac{\partial g}{\partial p}$$

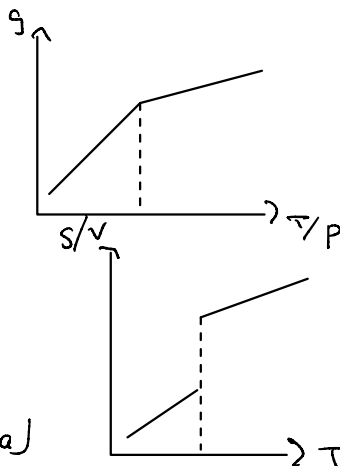
- Conditions

- s & $v \rightarrow$ change

- $\frac{\partial g}{\partial p}$ = discontinuous

(Ehrenfer classifier)

- latent heat present (Morder criteria)



Date:

$$C_p = \left(\frac{dq}{dT} \right)_p = T \left(\frac{\partial s}{\partial T} \right)_p \rightarrow \infty$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p \rightarrow \infty$$

$$k_T = - \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \rightarrow \infty$$

} because of
finite discontinuities

Clausius - Clapeyron relation

$$s = s(T, v)$$

$$ds = \left(\frac{\partial s}{\partial T} \right)_v dT + \left(\frac{\partial s}{\partial v} \right)_T dv$$

@ boundary $dT = 0$

$$ds = \left(\frac{\partial s}{\partial v} \right)_T dv$$

$$= \left(\frac{\partial p}{\partial T} \right)_T dv$$

$$\frac{\Delta p}{\Delta T} = \frac{ds}{dv} = \frac{s^p - s^i}{v^p - v^i}$$

$$T ds = L/dh$$

$$= \frac{dh}{T dv} = \frac{\Delta h}{T \Delta v} = \frac{L}{T \Delta v}$$

Date:

INTERNAL ENERGY EQUATIONS

$$dU = TdS - PdV$$

$$\left(\frac{dU}{dV}\right)_T = T\left(\frac{dS}{dV}\right)_T - P$$
$$= T\left(\frac{\partial P}{\partial T}\right)_V - P$$

Can do same with dP

KINETIC THEORY OF GAS

• Assumptions

- no interaction between particles
- all particle collision is elastic
- number of particles $\gg$ (statistical arguments apply)
 $\sim$ known as thermodynamic limit
- particle size $\ll$ interatomic distance
- used to explain transport properties

Date:

VELOCITY DISTRIBUTION

• Assume all particle is in equilibrium

$$P(\vec{v}) = e^{\underbrace{-(mv^2/2k_B T)}_{\text{Maxwell-Boltzmann factor}}} = e^{-E/k_B T} = e^{-\beta E}$$

• E' : E of reservoir ; E_r : E of micro state

$$E_0 = E^0 - E'$$

PROOF

• In interval E^0 to $E^0 + \delta E$, P to find particle with E_r

$$P_i \propto \Omega(E')$$

↳ set of all accessible microstate

$$E_r \ll E'$$

$$\ln(\Omega(E^0 - E_r)) \approx \ln(\Omega(E_0)) \left. \frac{d \ln(\Omega)}{d \ln E} \right|_{E_0} E_r \dots \rightarrow \text{negligible}$$

$$= \ln(\Omega[E_0]) - \beta E_r$$

$$\ln\left(\frac{\Omega(E^0 - E_r)}{\Omega[E^0]}\right) = -\beta E_r$$

$$\Rightarrow \Omega' = \Omega' e^{-\beta E_r}$$

$$P_r = c e^{-\beta E_r}$$

$g(v_x) = A e^{-mv^2/2k_B T} \rightarrow$ Distribution function

$$\int g(v) dv = n = N/V \Rightarrow A = n \left(\frac{m}{2k_B T \pi} \right)^{1/2}$$

Date:

Probability of finding particle in $(\vec{r}$ to $\vec{r} + d\vec{r}$ & $\vec{p}$ to $d\vec{p}$)

$$P(\vec{r}, \vec{p}) d^3\vec{r} d^3\vec{p} \propto e^{-\beta k} d^3\vec{r} d^3\vec{p}$$

$$\vec{v} = \vec{p}/m$$

$$N P(\vec{r}, \vec{p}) = f(\vec{r}, \vec{v})$$

$$f(\vec{r}, \vec{v}) = C e^{-\beta E} \quad - \text{Maxwell's velocity distribution}$$

$$\begin{aligned} N &= \iiint f(\vec{r}, \vec{v}) d^3\vec{r} d^3\vec{v} \\ &= C \int d^3\vec{r} \int e^{-\frac{mv^2}{2k_B T}} d^3\vec{v} \\ &= C V \int e^{-mv^2/2k_B T} d^3\vec{v} \end{aligned}$$

$$C = n \left(\frac{m}{2\pi k_B T} \right)^{3/2}$$

$f \rightarrow$ not explicitly dependent on $\vec{r}$

$f(\vec{r}, \vec{v}) d^3\vec{v}$ = Mean no. of particle/unit volume with velocity $\vec{v}$

$P \text{ for volume} = \frac{g(v_x)/n}{\text{odd fn}}$

$$\langle v_x \rangle = \frac{\int v_x g(x) d^3\vec{v}}{n} = 0$$

$$\langle v_x^2 \rangle = k_B T / m$$

$$v_{\text{rms}} = \sqrt{\frac{k_B T}{m}} \quad (1D)$$

